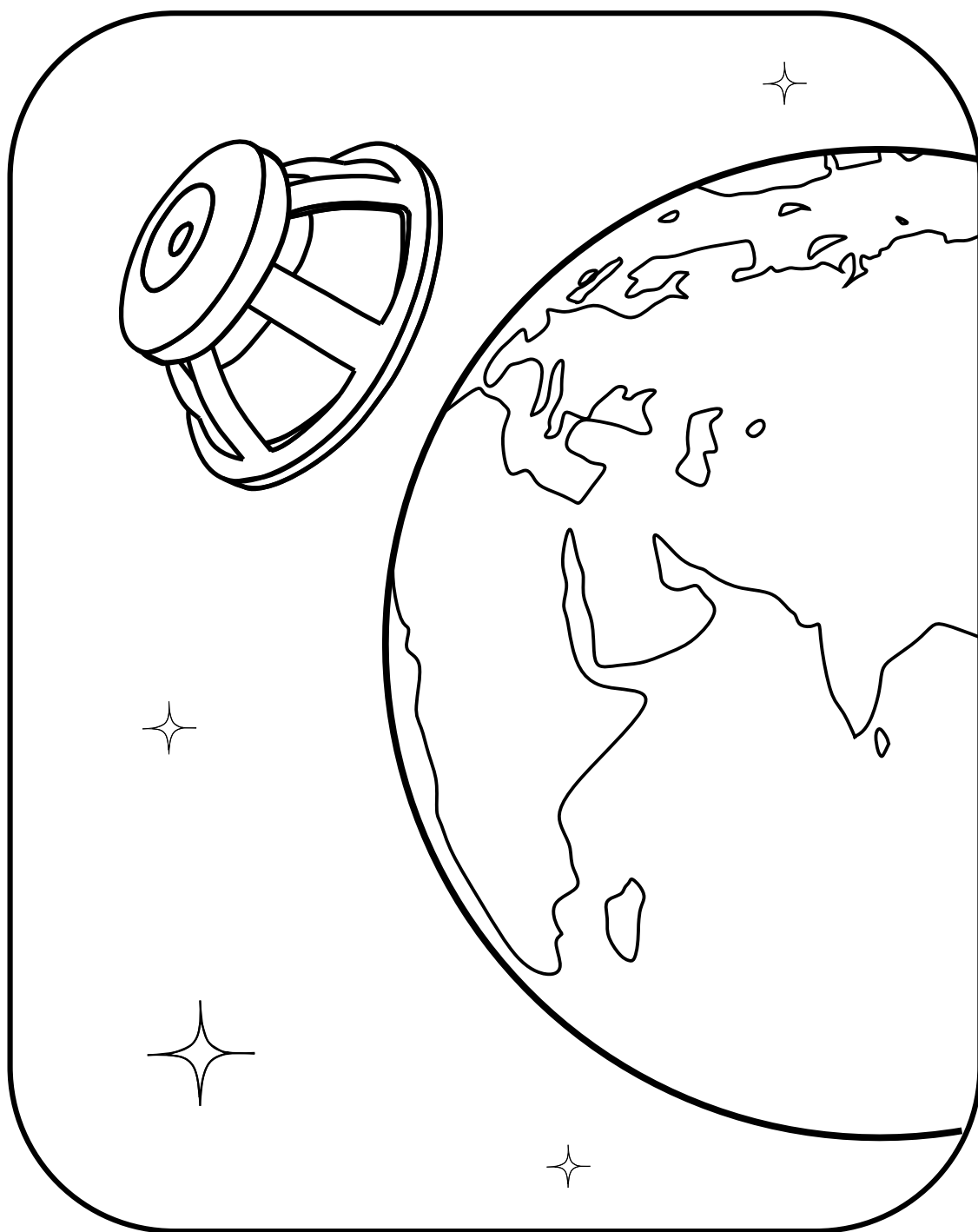


DEEP FIELD



AUDIO

The DEEP FIELD preamp is a project aiming at giving a starting point to those who want to step into building their own preamp. Even if almost all the knowledge needed to craft a preamp is available online, there is to my knowledge no full preamp projects or documentation.

The design and documentation are by Nélo Breuiller aka DEEP FIELD AUDIO.

Most of this project is based on the incredible work done by Rode Elliot on his website: <https://sound-au.com/> .

The parametric equaliser is a design by SENSITRONIC

Thanks to Arend from java hifi for building and troubleshooting this preamp

Thanks to the cooperative ouvrière de l'audio rennais for the support

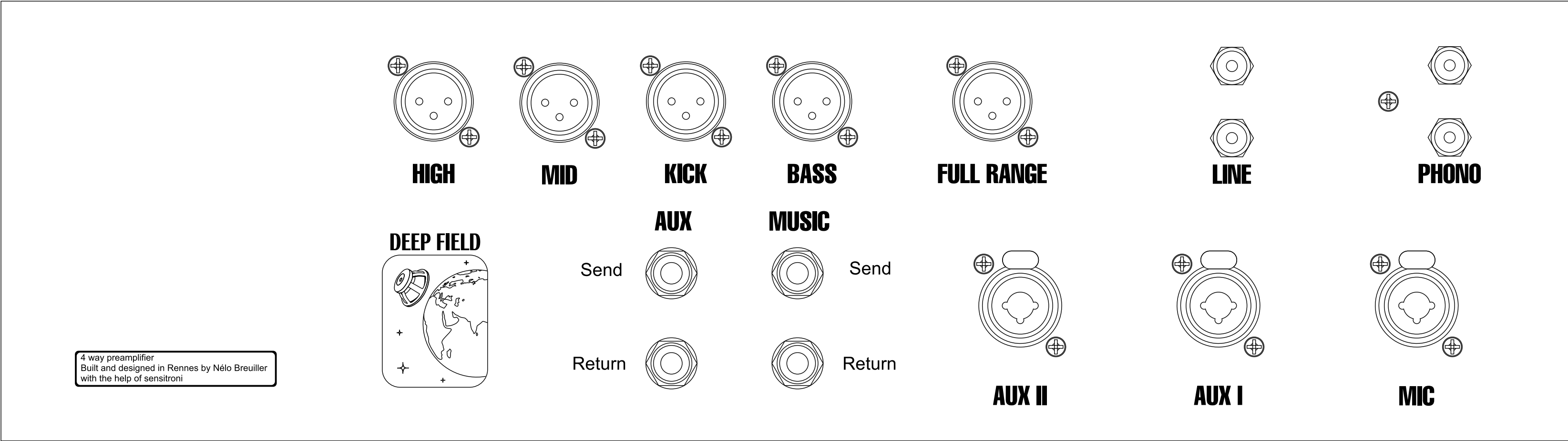
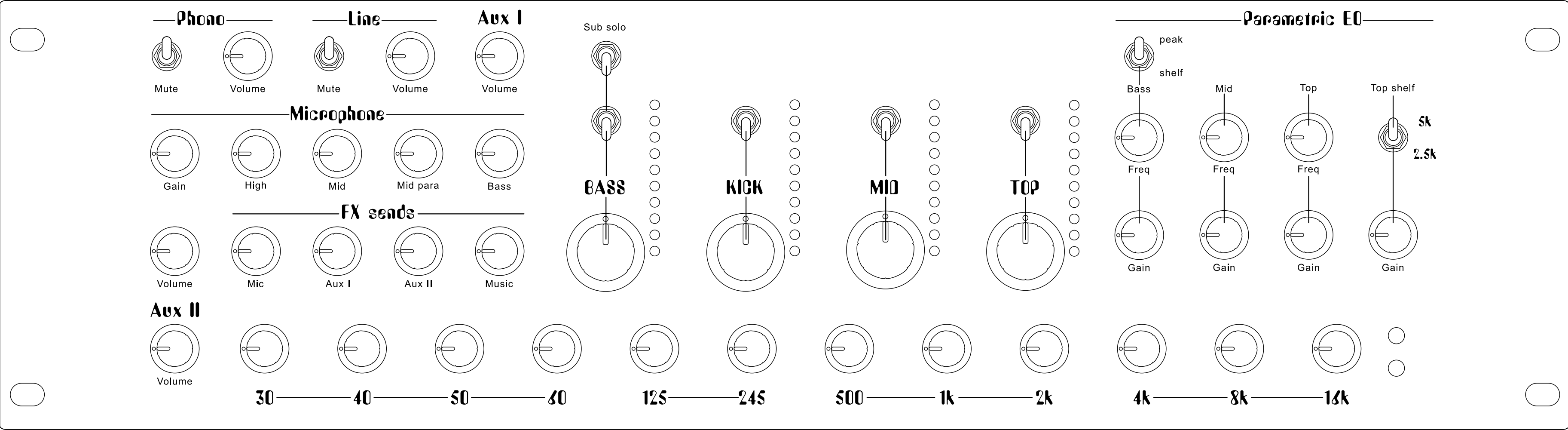
Thanks to the dubknowledge team for the knowledge gathering

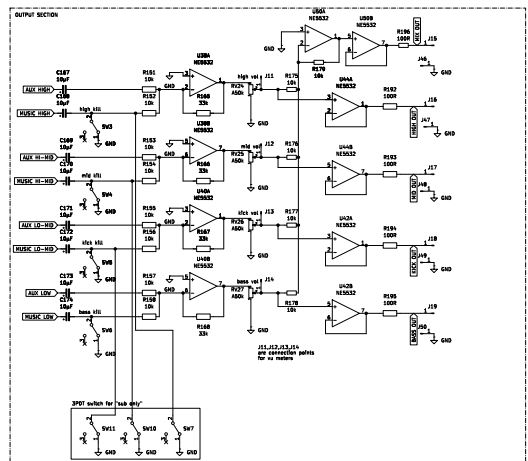
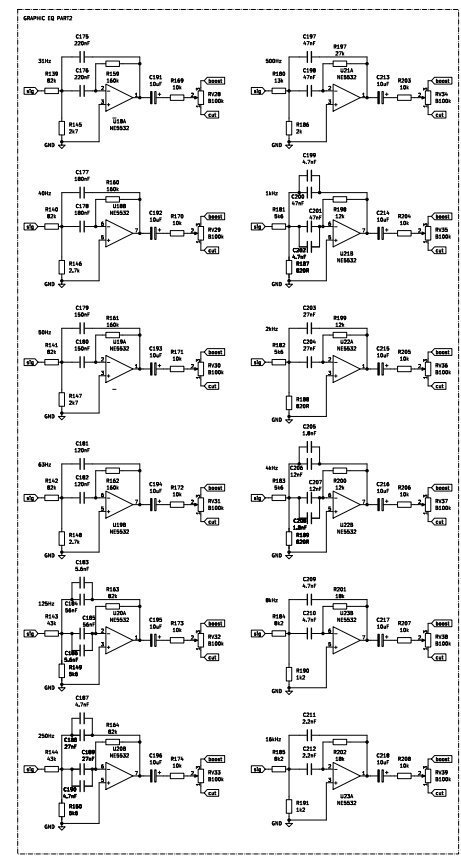
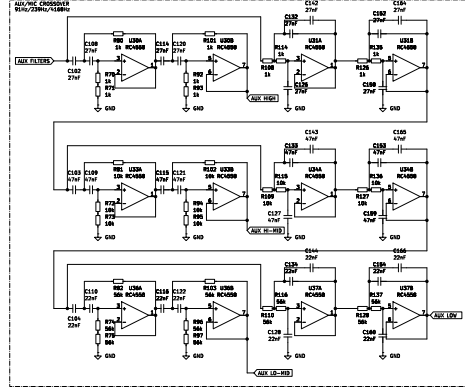
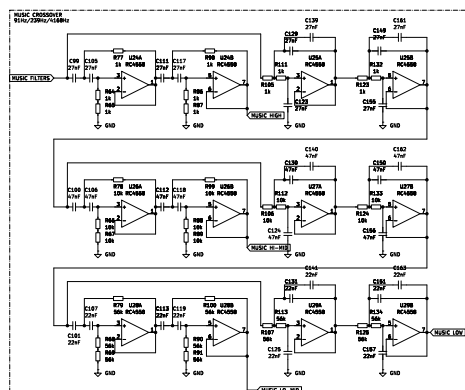
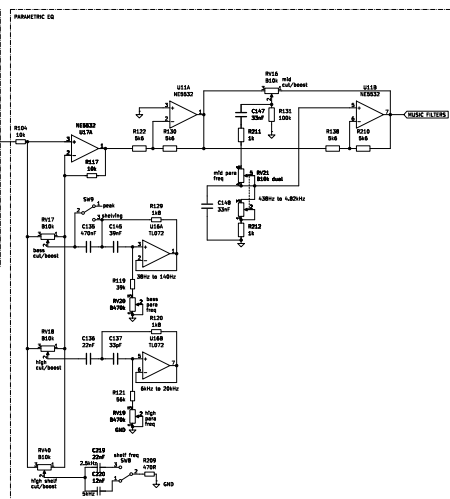
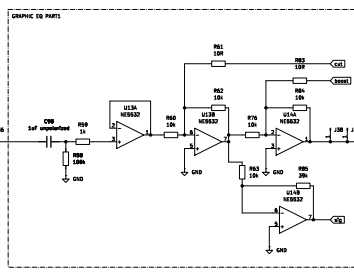
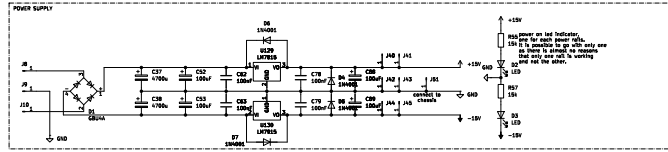
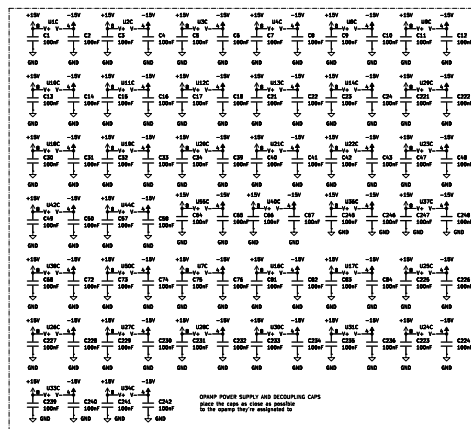
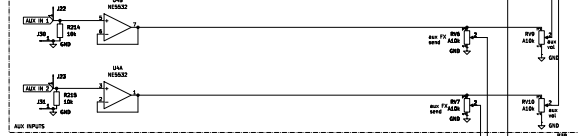
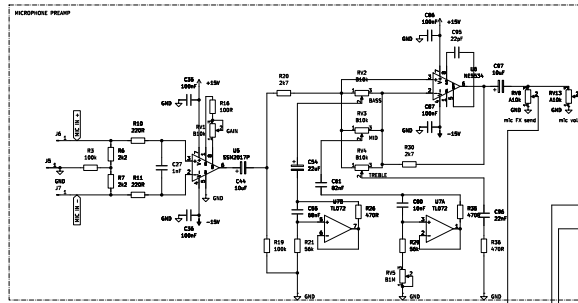
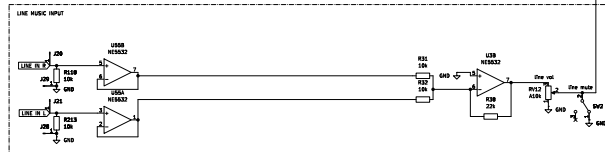
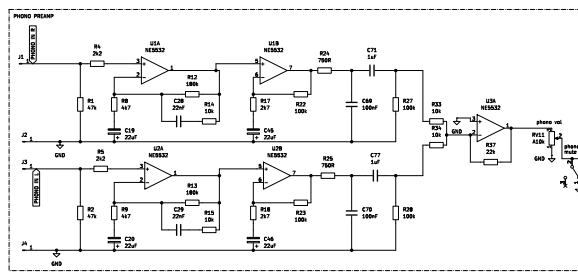
All the following documentation is open source and can be used for personal projects but using it for commercial purpose is strictly forbidden. Modifications and improvements are welcome, if you happen to make some revisions of this design, please share them under the same conditions.

I can provide the KiCad files to anyone who would like to make their own PCB design or schematic revisions. As this is the first version of this design, it is subject to mistakes and errors.

If you want to print your own PCB's I can provide you with the full ready to order gerber files

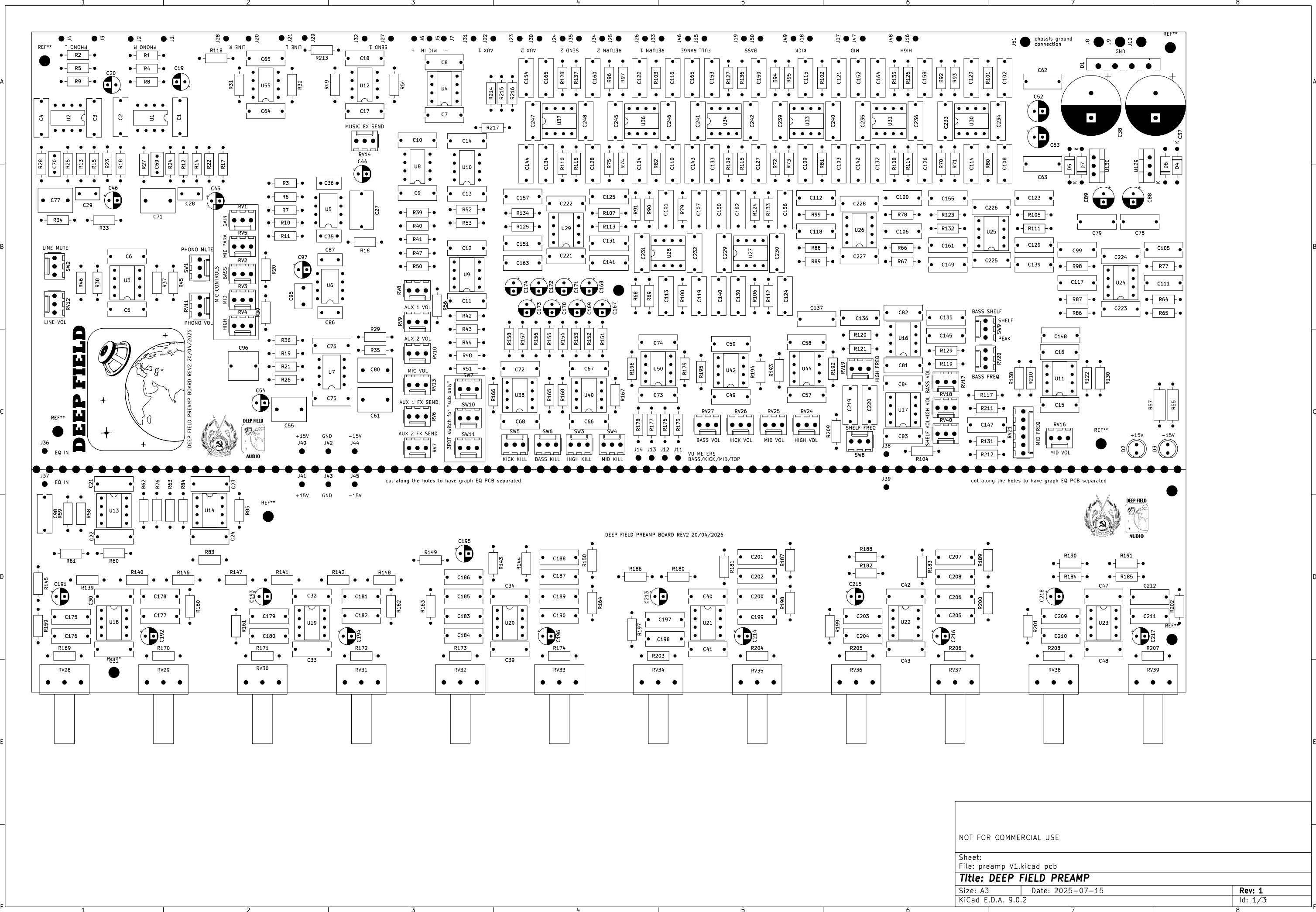
The DEEP FIELD is a 4-way preamplifier featuring separated crossover for music and aux/microphone, allowing music to be cut without cutting sirens, microphone or FXs. The music channel consists of one phono input and one line input both with mute switch and volume control. Every stereo input gets summed to mono. This channel also features one FX loop with send control, one 12 bands graphic EQ and a 3way parametric EQ stage plus top boost. The microphone input is balanced, it offers gain and volume control, a 3way EQ with parametric medium and a send control for FX. The two auxiliary inputs have volume control and send control for FX. Both microphone and aux inputs share the same FX loop. The crossover section offers volume control and kills switches for each way plus a sub only kill switch, and one (optional) ten led VU meter for each way. There is one output for each way (bass, kick, mid and top) and a full range output.





Design by NED BREUILLER
parametric EQ design by SENECHON
based on design by ROSE ELIOT
<https://sound-au.com/projects.htm>

Sheet 1 /
File: param_v1.14a2.sch
Title: DEEP FIELD PREAMP
Rev A1, 20th Jan 2024-01-15
R001 EDA 9.0.2



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Title: DEEP FIELD PREAMP

Size: A3 Date: 2025-07-15

KiCad E.D.A. 9.0.2

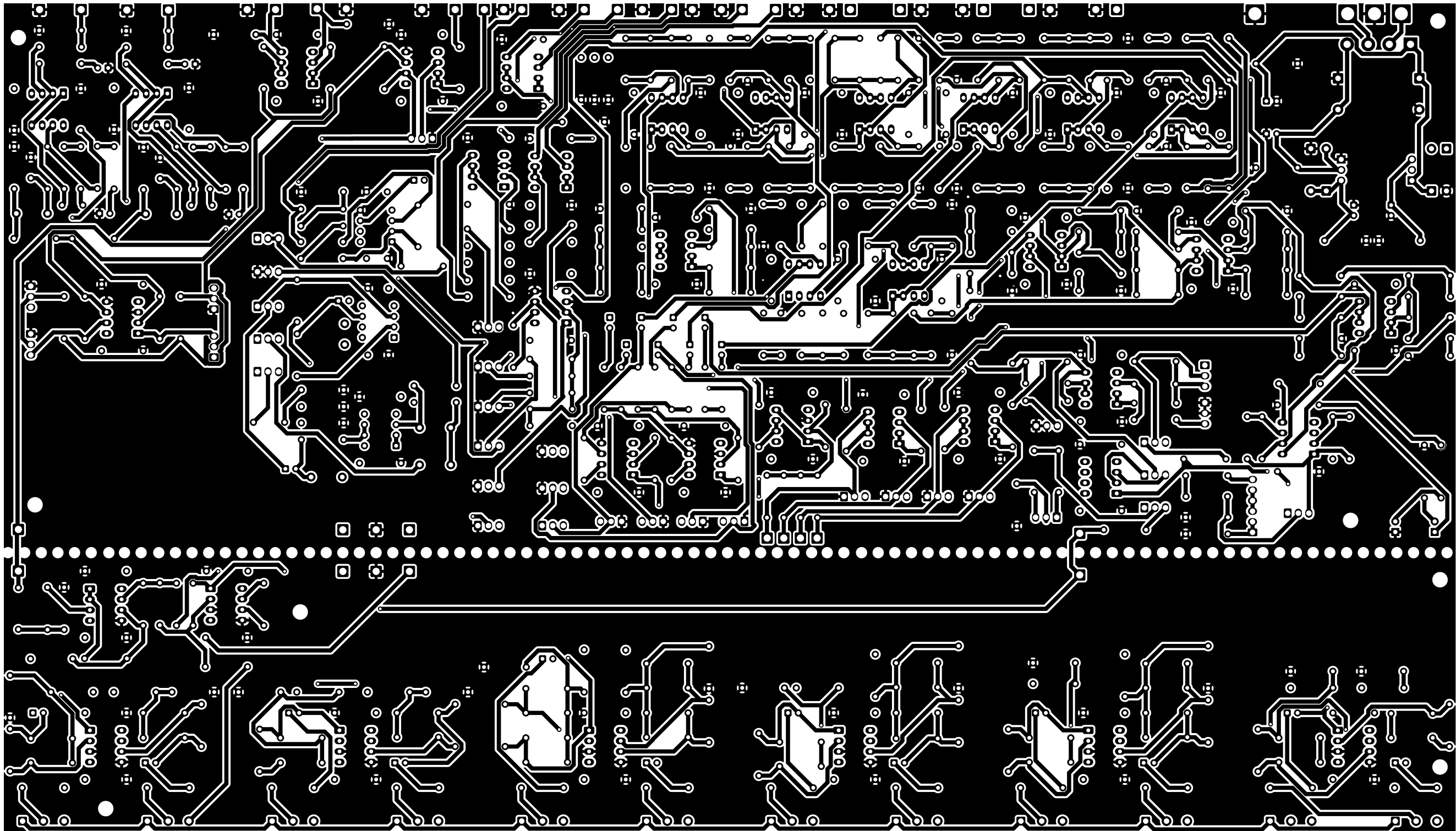
Rev: 1

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C19,C20,C45,C46,C54	22uF	5
C27	1nF	1
C28,C29,C96,C101,C104,C107,C110,C113,C116,C119,C122,C125,C128,C131,C134,C141,C144,C151,C154,C157,C160,C163,C166,C219	22nF	24
C37,C38	4700u	2
C44,C97,C191,C192,C193,C194,C195,C196,C213,C214,C215,C216,C217,C218	10uF	14
C52,C53,C88,C89	100uF	4
C55	68nF	1
C61	82nF	1
C71,C77	1uF	2
C80	10nF	1
C95	22pF	1
C98	1uF unpolarized	1
C99,C102,C105,C108,C111,C114,C117,C120,C123,C126,C129,C132,C139,C142,C149,C152,C155,C158,C161,C164,C188,C189,C203,C204	27nF	24
C100,C103,C106,C109,C112,C115,C118,C121,C124,C127,C130,C133,C140,C143,C150,C153,C156,C159,C162,C165,C197,C198,C200,C201	47nF	24
C135	470nF	1
C136,C206,C207,C220	12nF	4
C137	33pF	1
C145	39nF	1
C147,C148	33nF	2
C167,C168,C169,C170,C171,C172,C173,C174	10µF	8
C175,C176	220nF	2
C177,C178	180nF	2
C179,C180	150nF	2
C181,C182	120nF	2
C183,C186	5.6nF	2
C184,C185	56nF	2
C187,C190,C199,C202,C209,C210	4.7nF	6
C205,C208	1.8nF	2
C211,C212	2.2nF	2

D1	GBU4A	1
D2,D3	LED	2
R1,R2	47K	2
R3,R19,R22,R23,R27,R28,R58,R131	100k	8
R4,R5,R6,R7	2k2	4
R8,R9	4k7	2
R10,R11	220R	2
R12,R13	180k	2
R14,R15,R31,R32,R33,R34,R37,R38,R39,R40,R41,R42,R45,R46,R47,R48,R49,R51,R52,R53,R56,R60,R62,R63,R66,R67,R72,R73,R76,R78 R81,R84,R88,R89,R94,R95,R99,R102,R106,R109,R112,R115,R124,R127,R133,R136,R151,R152,R153,R154,R155,R156,R157,R158,R165 R166,R167,R168,R169,R170,R171,R172,R173,R174,R175,R176,R177,R178,R179,R203,R204,R205,R206,R207,R208	10k	75
R16,R50,R54,R192,R193,R194,R195,R196	100R	8
R17,R18,R20,R30,R104,R117,R145,R147	2k7	8
R21,R29,R68,R69,R74,R75,R79,R82,R90,R91,R96,R97,R100,R103,R107,R110,R113,R116,R121,R125,R128,R134,R137	56k	23
R24,R25	750R	2
R26,R35,R36,R209	470R	4
R43,R44	5k	2
R55,R57	15k	2
R59,R64,R65,R70,R71,R77,R80,R86,R87,R92,R93,R98,R101,R105,R108,R111,R114,R123,R126,R132,R135,R211,R212	1k	23
R61,R83	10R	2
R85,R119	39k	2
R120	270R	1
R122,R130,R138,R149,R150,R181,R182,R183,R210	5k6	9
R129	1k8	1
R139,R140,R141,R142,R163,R164	82k	6
R143,R144	43k	2
R146,R148	2.7k	2
R159,R160,R161,R162	160k	4
R180	13k	1
R184,R185	8.2k	2
R186	2k	1
R187,R188,R189	820R	3
R190,R191	1.2k	2

R197	27k	1
R198,R199,R200	12k	3
R201,R202	18k	2
RV1,RV2,RV3,RV4,RV6,RV7,RV8,RV9,RV10,RV11,RV12,RV13,RV14,RV16,RV21	10k	15
RV5	1M	1
RV17,RV20	470k	2
RV18	500k	1
RV19,RV28,RV29,RV30,RV31,RV32,RV33,RV34,RV35,RV36,RV37,RV38,RV39,RV40	100k	14
RV24,RV25,RV26,RV27	50k	4
SW1,SW2,SW3,SW4,SW5,SW6,SW7,SW8,SW9,SW10,SW11	SW_SPDT	11
U1,U2,U3,U4,U8,U9,U10,U11,U12,U13,U14,U17,U18,U19,U20,U21,U22,U23,U38,U40,U42,U44,U50,U55	NE5532	24
U5	SSM2017P	1
U6	NE5534	1
U7,U16	TL072	2
U24,U25,U26,U27,U28,U29,U30,U31,U33,U34,U36,U37	RC4558	12
U129	LM7815	1
U130	LM7915	1
transformer 18V-18V or 15V-15V 15VA minimum		1



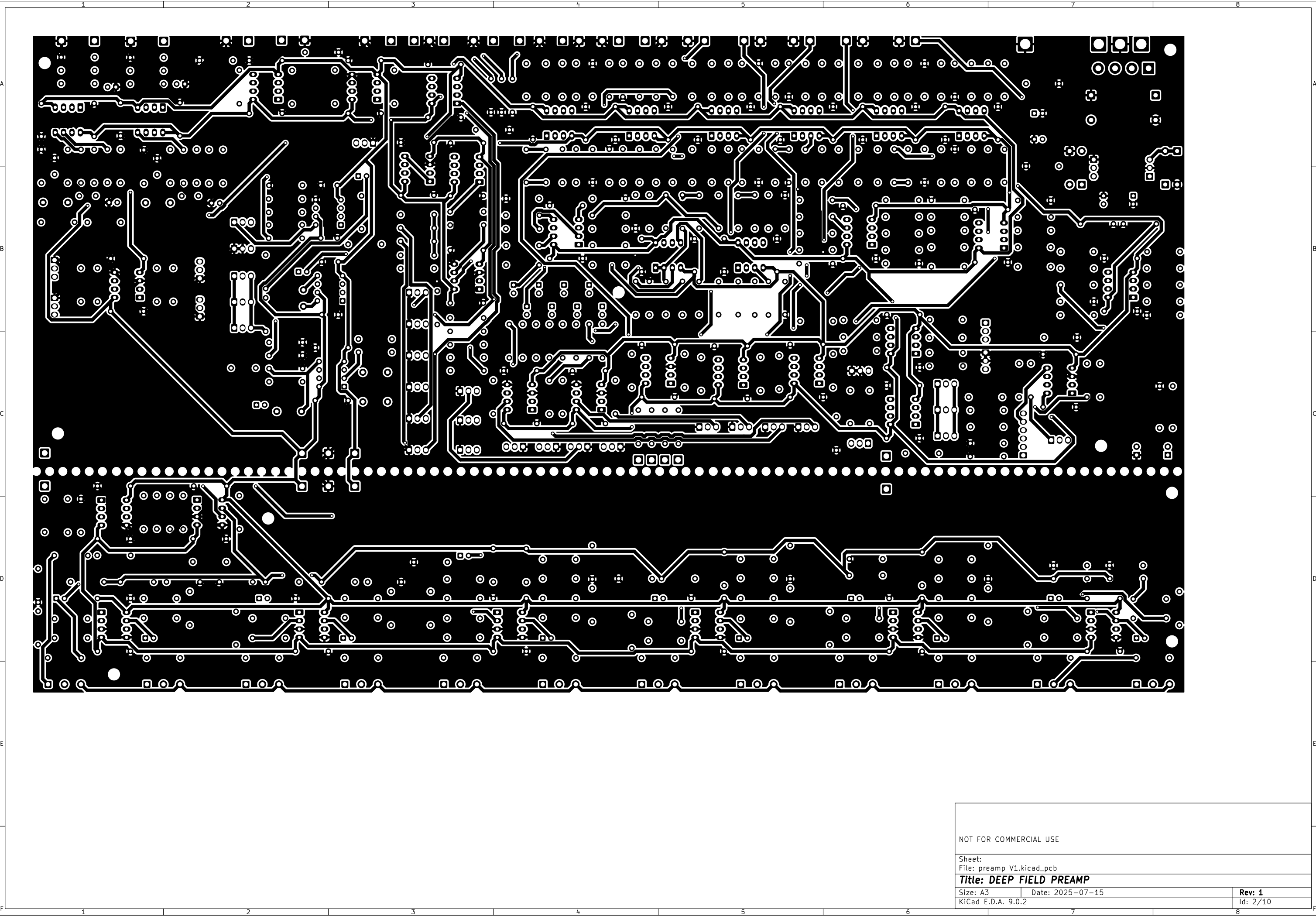
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KiCad E.D.A. 9.0.2

Rev: 1
Id: 1/10



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KiCad E.D.A. 9.0.2

Rev: 1
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